

习题 8.2

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- (5) 提示

$$\int \left(\frac{1}{\sqrt{3-x^2}} + \frac{1}{\sqrt{1-3x^2}} \right) dx = \int \frac{1}{\sqrt{3-x^2}} dx + \int \frac{1}{\sqrt{1-3x^2}} dx$$

分别令

$$x = \sqrt{3} \sin t$$

$$x = \frac{1}{\sqrt{3}} \sin t$$

- (12)

$$\begin{aligned} \int \frac{1}{1+\sin x} dx &= \int \frac{1-\sin x}{1-\sin^2 x} dx \\ &= \int \frac{1-\sin x}{\cos^2 x} dx \\ &= \int \frac{1}{\cos^2 x} dx - \int \frac{\sin x}{\cos^2 x} dx \\ &= \int \frac{1}{\cos^2 x} dx + \int \frac{1}{\cos^2 x} d(\cos x) \\ &= \tan x - \cos^{-1} x + C \\ &= \tan x - \sec x + C \end{aligned}$$

• (13)

$$\begin{aligned}\int \csc x dx &= \int \frac{\csc x (\cot x - \csc x)}{\cot x - \csc x} dx \\ &= \int \frac{d(\cot x - \csc x)}{\cot x - \csc x} \\ &= \ln |\cot x - \csc x| + C\end{aligned}$$

• (15)

$$\begin{aligned}\int \frac{x}{4+x^4} dx &= \frac{1}{4} \int \frac{x}{1+(\frac{x^2}{2})^2} \\ &= \frac{1}{4} \int \frac{1}{1+(\frac{x^2}{2})^2} d(\frac{x^2}{2}) \\ &= \frac{1}{4} \tan \frac{x^2}{2} + C\end{aligned}$$

• (19)

$$\begin{aligned}\int \frac{dx}{x(1+x)} &= \int \frac{dx}{x} - \int \frac{dx}{1+x} \\ &= \ln |x| - \ln |1+x| + C \\ &= \ln \left| \frac{x}{1+x} \right| + C\end{aligned}$$

• (21)

提示:

$$\int \cos^5 x dx = \int (1 - \sin^2 x)^2 d(\sin x)$$

• (22)

$$\begin{aligned}\int \frac{dx}{\sin x \cos x} &= \int \frac{dx}{\tan x \cos^2 x} \\ &= \int \frac{d(\tan x)}{\tan x} \\ &= \ln |\tan x| + C\end{aligned}$$

• (23)

$$\begin{aligned}
 \int \frac{dx}{e^x + e^{-x}} &= \int \frac{dx}{\frac{(e^x)^2 + 1}{e^x}} \\
 &= \int \frac{e^x}{(e^x)^2 + 1} dx \\
 &= \int \frac{1}{(e^x)^2 + 1} d(e^x) \\
 &= \arctan e^x + C
 \end{aligned}$$

• (25)

$$\begin{aligned}
 \int \frac{x^2 + 2}{(x + 1)^3} dx &= \int \frac{(x + 1)^2 - 2(x + 1) + 3}{(x + 1)^3} dx \\
 &= \int \frac{(x + 1)^2 - 2(x + 1) + 3}{(x + 1)^3} d(x + 1) \\
 &= \int \frac{1}{x + 1} d(x + 1) - 2 \int \frac{1}{(x + 1)^2} d(x + 1) + 3 \int \frac{1}{(x + 1)^3} d(x + 1) \\
 &= \ln |x + 1| + 2(x + 1)^{-1} - \frac{3}{2}(x + 1)^{-2} + C
 \end{aligned}$$

• (26)

提示：令 $x = atant$

• (29)

令 $x = t^6 (t \geq 0, t \neq 1), dx = 6t^5 dt$ 。

$$\begin{aligned}
 \int \frac{\sqrt{x}}{1 - \sqrt[3]{x}} dx &= \int \frac{t^3}{1 - t^2} 6t^5 dt \\
 &= 6 \int \frac{t^8}{1 - t^2} dt \\
 &= 6 \int \frac{t^8 - 1 + 1}{1 - t^2} dt
 \end{aligned}$$

我们有

$$\begin{aligned}
 t^8 - 1 &= (t^2)^4 - (1^2)^4 \\
 &= (t^2 - 1)[(t^2)^3 + (t^2)^2 + t^2 + 1]
 \end{aligned}$$

所以

$$\begin{aligned}
 6 \int \frac{t^8 - 1 + 1}{1 - t^2} dt &= 6 \int -t^6 - t^4 - t^2 - 1 + \frac{1}{1 - t^2} dt \\
 &= 6 \left(-\frac{1}{7}t^7 - \frac{1}{5}t^5 - \frac{1}{3}t^3 - t + \frac{1}{2} \ln \left| \frac{t+1}{t-1} \right| \right) + C \\
 &= -\frac{6}{7}t^7 - \frac{6}{5}t^5 - 2t^3 - 6t + 3 \ln \left| \frac{t+1}{t-1} \right| + C \\
 &= -\frac{6}{7}x^{\frac{7}{6}} - \frac{6}{5}x^{\frac{5}{6}} - 2x^{\frac{1}{2}} - 6x^{\frac{1}{6}} + 3 \left| \frac{x^{\frac{1}{6}} + 1}{x^{\frac{1}{6}} - 1} \right| + C
 \end{aligned}$$

• (30)

令 $t = \sqrt{x+1}$, $x = t^2 - 1$, $dx = 2t dt$ 。

$$\begin{aligned}
 \int \frac{\sqrt{x+1} - 1}{\sqrt{x+1} + 1} dx &= \int \frac{t-1}{t+1} 2t dt \\
 &= 2 \int \frac{(t-1)t}{t+1} dt
 \end{aligned}$$

说明 1. 我们有 $t+a$, 使得

$$\begin{aligned}
 (t+1)(t+a) + b &= (t-1)t \\
 t^2 + at + t + a + b &= t^2 - t \\
 t^2 + (a+1)t + a + b &= t^2 - t
 \end{aligned}$$

于是

$$\begin{cases} a+1 = -1 \\ a+b = 0 \end{cases}$$

可得

$$a = -2, b = 2$$

于是

$$(t+1)(t-2) + 2 = (t-1)t$$

所以

$$\begin{aligned} 2 \int \frac{(t-1)t}{t+1} dt &= 2 \int \frac{(t+1)(t-2)+2}{t+1} dt \\ &= 2 \int t-2 + \frac{2}{t+1} dt \\ &= 2\left(\frac{1}{2}t^2 - 2t + 2\ln|t+1|\right) + C \\ &= t^2 - 4t + 4\ln|t+1| + C \end{aligned}$$

剩下的部分，略

• (31)

令 $t = 1 - 2x, x = \frac{1-t}{2}, dx = -\frac{1}{2}dt$ 。

$$\begin{aligned} \int x(1-2x)^{99} dx &= -\frac{1}{2} \int \frac{1-t}{2} t^{99} dt \\ &= -\frac{1}{4} \int (1-t)t^{99} dt \\ &= -\frac{1}{4} \int t^{99} - t^{100} dt \\ &= -\frac{1}{4} \left(\frac{1}{100} t^{100} - \frac{1}{101} t^{101} \right) + C \\ &= -\frac{1}{400} (1-2x)^{100} + \frac{1}{404} (1-2x)^{101} + C \end{aligned}$$

• (32)

提示：

$$\frac{1}{x(1+x^n)} = \frac{1}{x} - \frac{x^{n-1}}{1+x^n}$$

• (33)

$$\begin{aligned} \int \frac{x^{2n-1}}{x^n+1} dx &= \frac{1}{n} \int \frac{x^n}{x^n+1} d(x^n) \\ &= \frac{1}{n} \int 1 - \frac{1}{x^n+1} d(x^n) \\ &= \frac{1}{n} (x^n - \ln|x^n+1|) + C \end{aligned}$$

• (35)

$$\begin{aligned}
 \int \frac{\ln 2x}{x \ln 4x} dx &= \int \frac{\ln 4x - \ln 2}{x \ln 4x} dx \\
 &= \int \frac{1}{x} - \frac{\ln 2}{x \ln 4x} dx \\
 &= \int \frac{1}{x} dx - \ln 2 \int \frac{1}{x \ln 4x} dx \\
 &= \int \frac{1}{x} dx - \ln 2 \int \frac{1}{\ln 4x} d(\ln 4x) \\
 &= \ln x - \ln 2 \cdot \ln |\ln 4x| + C
 \end{aligned}$$

• (36)

令 $x = \sec t, dx = \sec t \cdot \tan t dt$ 。【这里的 $t \in (0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2})$ ，保证 $\sqrt{\tan^2 t}$ 的绝对值可以去掉。考试时可以不写这部分】

$$\begin{aligned}
 \int \frac{dx}{x^4 \sqrt{x^2 - 1}} &= \int \frac{\sec t \cdot \tan t}{\sec^4 x \cdot \tan t} dt \\
 &= \int \frac{1}{\sec^3 x} dt \\
 &= \int \cos^3 t dt \\
 &= \int (1 - \sin^2 t) d(\sin t) \\
 &= \sin t - \frac{1}{3} \sin^3 t + C
 \end{aligned}$$

然后通过三角图形，把 t 换成 x 即可。

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• (1)

$$\begin{aligned}
 I_n &= \int \tan^n x dx \\
 &= \int \tan^{n-2} x \tan^2 x dx \\
 &= \int \tan^{n-2} x (\sec^2 x - 1) dx \\
 &= \int \tan^{n-2} x \sec^2 x dx - \int \tan^{n-2} x dx \\
 &= \int \tan^{n-2} x d(\tan x) - I_{n-2} \\
 &= \frac{1}{n-1} \tan^{n-1} x - I_{n-2}
 \end{aligned}$$

注:

$$\begin{aligned}
 I_0 &= \int \tan^0 x dx = \int 1 dx = x + C \\
 I_1 &= \int \tan x dx = -\ln |\cos x| + C
 \end{aligned}$$

• (2)

$$\begin{aligned}
 I(m, n) &= \int \cos^m x \sin^n x dx \\
 &= \int \cos^{m-2} x \cos^2 x \sin^n x dx \\
 &= \int \cos^{m-2} x (1 - \sin^2 x) \sin^n x dx \\
 &= \int \cos^{m-2} x \sin^n x dx - \int \cos^{m-2} x \sin^{n+2} x dx \\
 &= I(m-2, n) + \frac{1}{m-1} \int \sin^{n+1} x d(\cos^{m-1} x) \\
 &= I(m-2, n) + \frac{1}{m-1} (\cos^{m-1} x \sin^{n+1} - \int \cos^m x \sin^n x dx) \\
 &= I(m-2, n) + \frac{1}{m-1} (\cos^{m-1} x \sin^{n+1} - I(m, n)) \\
 &= I(m-2, n) + \frac{1}{m-1} \cos^{m-1} x \sin^{n+1} - \frac{1}{m-1} I(m, n)
 \end{aligned}$$

所以

$$\begin{aligned}
 I(m, n) + \frac{1}{m-1}I(m, n) &= I(m-2, n) + \frac{1}{m-1}\cos^{m-1}x\sin^{n+1} \\
 \frac{m+n}{m-1}I(m, n) &= I(m-2, n) + \frac{1}{m-1}\cos^{m-1}x\sin^{n+1} \\
 I(m, n) &= \frac{1}{m+n}\cos^{m-1}x\sin^{n+1} + \frac{m-1}{m+n}I(m-2, n)
 \end{aligned}$$

类似地，可证另一个等式。

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- (1)

$$\begin{aligned}
 I_n &= \int x^n e^{kx} dx \\
 &= \frac{1}{k} \int x^n d(e^{kx}) \\
 &= \frac{1}{k} (x^n e^{kx} - n \int x^{n-1} e^{kx} dx) \\
 &= \frac{1}{k} x^n e^{kx} - \frac{n}{k} I_{n-1}
 \end{aligned}$$

- (2)

$$\begin{aligned}
 I_n &= \int (\ln x)^n dx \\
 &= \int (\ln x)^n d(x) \\
 &= x(\ln x)^n - \int x \cdot \frac{n}{x} (\ln x)^{n-1} dx \\
 &= x(\ln x)^n - n \int (\ln x)^{n-1} dx \\
 &= x(\ln x)^n - n I_{n-1}
 \end{aligned}$$

• (3)

$$\begin{aligned}
I_n &= \int (\arcsin x)^n dx \\
&= x(\arcsin x)^n - \int x \cdot n(\arcsin x)^{n-1} \frac{1}{\sqrt{1+x^2}} dx \\
&= x(\arcsin x)^n + n \int (\arcsin x)^{n-1} d(\sqrt{1+x^2}) \\
&= x(\arcsin x)^n + n \left[\sqrt{1+x^2} (\arcsin x)^{n-1} - \int \sqrt{1+x^2} (n-1) (\arcsin x)^{n-2} \frac{1}{\sqrt{1+x^2}} dx \right] \\
&= x(\arcsin x)^n + n \left[\sqrt{1+x^2} (\arcsin x)^{n-1} - (n-1) \int (\arcsin x)^{n-2} dx \right] \\
&= x(\arcsin x)^n + n \left[\sqrt{1+x^2} (\arcsin x)^{n-1} - (n-1) I_{n-2} \right] \\
&= x(\arcsin x)^n + n \sqrt{1+x^2} (\arcsin x)^{n-1} - n(n-1) I_{n-2}
\end{aligned}$$

• (4)

$$\begin{aligned}
I_n &= \int e^{ax} \sin^n x dx \\
&= \frac{1}{a} \int \sin^n x d(e^{ax}) \\
&= \frac{1}{a} \left[e^{ax} \sin^n x - n \int e^{ax} \sin^{n-1} x \cos x dx \right] \\
&= \frac{1}{a} \left[e^{ax} \sin^n x - \frac{n}{a} \int \sin^{n-1} x \cos x d(e^{ax}) \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} \left[e^{ax} \sin^{n-1} x \cos x - \int e^{ax} ((n-1) \sin^{n-2} x \cos^2 x + \sin^{n-1} x (-\sin x)) dx \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} \left[e^{ax} \sin^{n-1} x \cos x - \int e^{ax} ((n-1) \sin^{n-2} x \cos^2 x - \sin^n x) dx \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} \left[e^{ax} \sin^{n-1} x \cos x - (n-1) \int e^{ax} \sin^{n-2} x \cos^2 x dx + \int e^{ax} \sin^n x dx \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} \left[e^{ax} \sin^{n-1} x \cos x - (n-1) \int e^{ax} \sin^{n-2} x (1 - \sin^2 x) dx + I_n \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} \left[e^{ax} \sin^{n-1} x \cos x - (n-1) \left(\int e^{ax} \sin^{n-2} x dx - \int e^{ax} \sin^n x dx \right) + I_n \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} \left[e^{ax} \sin^{n-1} x \cos x - (n-1)(I_{n-2} - I_n) + I_n \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} \left[e^{ax} \sin^{n-1} x \cos x - (n-1)I_{n-2} + nI_n \right] \\
&= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} e^{ax} \sin^{n-1} x \cos x + \frac{n(n-1)}{a^2} I_{n-2} - \frac{n^2}{a^2} I_n
\end{aligned}$$

所以

$$\begin{aligned}
(1 + \frac{n^2}{a^2})I_n &= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} e^{ax} \sin^{n-1} x \cos x + \frac{n(n-1)}{a^2} I_{n-2} \\
\frac{n^2 + a^2}{a^2} I_n &= \frac{1}{a} e^{ax} \sin^n x - \frac{n}{a^2} e^{ax} \sin^{n-1} x \cos x + \frac{n(n-1)}{a^2} I_{n-2} \\
I_n &= \frac{1}{n^2 + a^2} (a e^{ax} \sin^n x - n e^{ax} \sin^{n-1} x \cos x + n(n-1) I_{n-2}) \\
I_n &= \frac{1}{n^2 + a^2} [e^{ax} \sin^{n-1} x (a \sin x - n \cos x) + n(n-1) I_{n-2}]
\end{aligned}$$